

IT PROFESSIONAL PORTFOLIO

Cloud Engineering and Security Projects

PROJECT 06

Create an EC2 IAM Role for AWS Systems Manager

AWS IAM | Amazon EC2 | AWS Systems Manager

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ENVIRONMENT	Personal AWS Lab
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Executive Summary

Created an AWS IAM role named **SSMInstance** for Amazon EC2. The role uses **AWS service** as the trusted entity type, selects **EC2** as the service use case, and uses the **EC2 Role for AWS Systems Manager** configuration. The role trusts the EC2 service to assume it and includes the AWS-managed **AmazonSSMManagedInstanceCore** permissions policy.

Business Problem

EC2 instances often need to call AWS services. Embedding long-term access keys on a virtual machine would create unnecessary credential risk. An IAM role provides a safer service identity model by allowing EC2 to receive temporary credentials and only the permissions attached to the role.

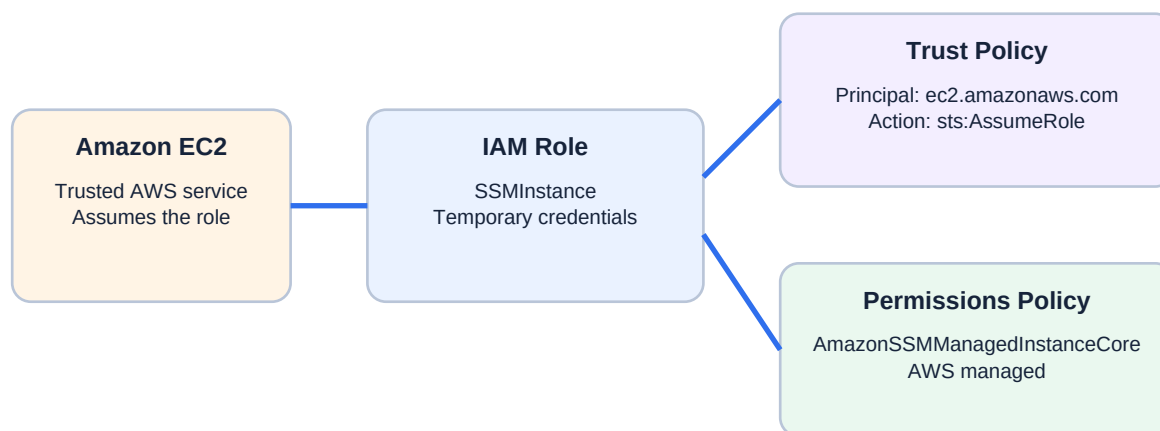
Objectives

- Create an IAM role for an AWS service.
- Configure Amazon EC2 as the trusted service.
- Select the EC2 Role for AWS Systems Manager use case.
- Attach the AmazonSSMManagedInstanceCore AWS-managed policy.
- Validate the trust policy that allows EC2 to assume the role.
- Prepare an instance role for future Systems Manager managed-instance projects.

Environment

Cloud Provider	Amazon Web Services (AWS)
Service	AWS Identity and Access Management (IAM)
Role Name	SSMInstance
Trusted Entity Type	AWS service
Trusted Service	Amazon EC2
Use Case	EC2 Role for AWS Systems Manager
Trust Principal	ec2.amazonaws.com
Assume Role Action	sts:AssumeRole
Permissions Policy	AmazonSSMManagedInstanceCore
Policy Type	AWS managed

Architecture



EC2 is trusted to assume the SSMInstance role, while the attached permissions policy defines what the temporary role session can do.

Figure 1 - EC2 IAM role trust and permissions architecture.

Implementation Summary

1. Opened IAM and started the Create role workflow.
2. Selected AWS service as the trusted entity type.
3. Selected EC2 as the service or use case.
4. Selected EC2 Role for AWS Systems Manager.
5. Reviewed the automatically selected AmazonSSMManagedInstanceCore policy.
6. Named the role SSMInstance.
7. Validated the trust policy that allows the EC2 service principal to call sts:AssumeRole.
8. Validated that AmazonSSMManagedInstanceCore was attached as the role's permissions policy.

KEY CONCEPT

The trust policy and permissions policy answer different questions: the trust policy defines **who or what can assume the role**, while the permissions policy defines **what the role is allowed to do after it is assumed**.

Evidence

Figure 2 - Trusted Entity and EC2 Systems Manager Use Case

Step 1: Select trusted entity

Step 2: Add permissions

Step 3: Name, review, and create

Select trusted entity Info

Trusted entity type

- ☒ **AWS service**
Allow AWS services like EC2, Lambda, or others to perform actions in this account.
- ☐ **AWS account**
Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.
- ☐ **Web identity**
Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.
- ☐ **SAML 2.0 federation**
Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.
- ☐ **Custom trust policy**
Create a custom trust policy to enable others to perform actions in this account.

Use case
Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case

EC2

Choose a use case for the specified service.

Use case

- ☐ **EC2**
Allows EC2 instances to call AWS services on your behalf.
- ☒ **EC2 Role for AWS Systems Manager**
Allows EC2 instances to call AWS services like CloudWatch and Systems Manager on your behalf.
- ☐ **EC2 Spot Fleet Role**
Allows EC2 Spot Fleet to request and terminate Spot Instances on your behalf.
- ☐ **EC2 - Spot Fleet Auto Scaling**
Allows Auto Scaling to access and update EC2 spot fleets on your behalf.
- ☐ **EC2 - Spot Fleet Tagging**
Allows EC2 to launch spot instances and attach tags to the launched instances on your behalf.
- ☐ **EC2 - Spot Instances**
Allows EC2 Spot Instances to launch and manage spot instances on your behalf.
- ☐ **EC2 - Spot Fleet**
Allows EC2 Spot Fleet to launch and manage spot fleet instances on your behalf.
- ☐ **EC2 - Scheduled Instances**
Allows EC2 Scheduled Instances to manage instances on your behalf.

The IAM role creation page shows AWS service selected as the trusted entity type, EC2 selected as the service, and EC2 Role for AWS Systems Manager selected as the use case.

Figure 3 - AmazonSSMManagedInstanceCore Permissions Policy

Add permissions Info

Permissions policies (1) Info

The type of role that you selected requires the following policy.

Policy name	Type
AmazonSSMManagedInstanceCore	AWS managed

► Set permissions boundary - optional

Cancel Previous Next

The Add permissions step shows the AWS-managed AmazonSSMManagedInstanceCore policy selected for the role.

Figure 4 - Role Name, Trust Policy, and Permission Assignment

Role details

Role name
Enter a meaningful name to identify this role.

Maximum 64 characters. Use alphanumeric and '+', '@', '-' characters.

Description
Add a short explanation for this role.

Maximum 1000 characters. Use letters (A-Z and a-z), numbers (0-9), tabs, new lines, or any of the following characters: '_', '@', '/', '!', '#', '\$', '%', '^', '&', '*', '~', '(', ')', '[', ']', '{', '}', '<', '>', '=', '"', ''', ':', ',', 'B;', 'D;', 'C;', '<', '>', '?', '@', 'A', 'B', 'C', 'D', 'E', 'F', 'G', 'H', 'I', 'J', 'K', 'L', 'M', 'N', 'O', 'P', 'Q', 'R', 'S', 'T', 'U', 'V', 'W', 'X', 'Y', 'Z', '[', '\', ']', '^', '_', '`', 'a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j', 'k', 'l', 'm', 'n', 'o', 'p', 'q', 'r', 's', 't', 'u', 'v', 'w', 'x', 'y', 'z', '{', '|', '}', '~', '', '€', '', '‚', 'ƒ', '„', '…', '†', '‡', 'ˆ', '‰', 'Š', '‹', 'Œ', '', 'Ž', '', '', '‘', '’', '“', '”', '•', '–', '—', '˜', '™', 'š', '›', 'œ', '', 'ž', 'Ÿ', ' ', '¡', '¢', '£', '¤', '¥', '¦', '§', '¨', '©', 'ª', '«', '¬', '­', '®', '¯', '°', '±', '²', '³', '´', 'µ', '¶', '·', '¸', '¹', 'º', '»', '¼', '½', '¾', '¿', 'À', 'Á', 'Â', 'Ã', 'Ä', 'Å', 'Æ', 'Ç', 'È', 'É', 'Ê', 'Ë', 'Ì', 'Í', 'Î', 'Ï', 'Ð', 'Ñ', 'Ò', 'Ó', 'Ô', 'Õ', 'Ö', '×', 'Ø', 'Ù', 'Ú', 'Û', 'Ü', 'Ý', 'Þ', 'ß', 'à', 'á', 'â', 'ã', 'ä', 'å', 'æ', 'ç', 'è', 'é', 'ê', 'ë', 'ì', 'í', 'î', 'ï', 'ð', 'ñ', 'ò', 'ó', 'ô', 'õ', 'ö', '÷', 'ø', 'ù', 'ú', 'û', 'ü', 'ý', 'þ', 'ÿ'.

Step 1: Select trusted entities Edit

Trust policy

```
1 - {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Sid": "",
6       "Effect": "Allow",
7       "Principal": {
8         "Service": "ec2.amazonaws.com"
9       },
10      "Action": "sts:AssumeRole"
11    }
12  ]
13 }
```

Step 2: Add permissions Edit

Permissions policy summary

Policy name	Type	Attached as
AmazonSSMManagedInstanceCore	AWS managed	Permissions policy

Step 3: Add tags

The review page shows the SSMInstance role name, a trust policy allowing the EC2 service principal to call sts:AssumeRole, and the AmazonSSMManagedInstanceCore permissions policy attached to the role.

Security Analysis

The role avoids placing static AWS access keys on an EC2 instance. When the role is used by an instance, AWS can provide temporary credentials for a limited session instead of permanent credentials.

The trust policy restricts role assumption to the EC2 service principal. This separates the question of who can use the role from the permissions the role grants.

The AmazonSSMManagedInstanceCore policy provides the standard permissions needed for a managed instance to communicate with AWS Systems Manager services.

The role is a foundation for centralized instance management and can reduce reliance on direct administrative access methods when later paired with Systems Manager capabilities.

Lessons Learned

AWS IAM roles are not the same as static user accounts. They are assumed to create sessions that use temporary credentials.

Every role has two core authorization components: a trust policy defining who or what can assume it, and one or more permissions policies defining what it can do.

Service roles allow AWS services such as EC2 to call other AWS services without embedding long-term credentials in the workload.

Production Improvements

- Attach the role only to EC2 instances that require Systems Manager functionality.
- Review the AWS-managed policy periodically and replace it with a narrower custom policy when the workload requires fewer capabilities.
- Use Systems Manager Session Manager and other managed access workflows to reduce dependence on inbound administrative ports.
- Monitor role assumption and related API activity through CloudTrail.
- Use tagging and naming standards to improve role ownership and lifecycle management.

Skills Demonstrated

IAM role creation	Temporary credentials	Amazon EC2
Trust policy review	Service identity design	AWS Systems Manager
Managed policy assignment	Static credential reduction	Instance role preparation

Resume Bullet

RESUME

Created an AWS IAM service role for EC2 with a trust policy allowing `ec2.amazonaws.com` to assume the role and attached `AmazonSSMManagedInstanceCore` permissions to prepare instances for secure AWS Systems Manager management.

STAR Interview Story

Situation: Future EC2 instances in the lab needed a secure way to communicate with AWS Systems Manager.

Task: Provide the required permissions without storing long-term AWS credentials on the instances.

Action: I created the `SSMInstance` IAM role, selected AWS service as the trusted entity type, chose EC2 and the Systems Manager use case, reviewed the trust policy, and attached `AmazonSSMManagedInstanceCore`.

Result: The environment now has a reusable EC2 service role that can provide temporary credentials and the permissions needed for future Systems Manager managed-instance workflows.

Version History

v1.0.0	July 9, 2026	Initial documented implementation. Created the SSMMInstance IAM role for the EC2 service, selected the EC2 Role for AWS Systems Manager use case, attached AmazonSSMMManagedInstanceCore, validated the EC2 trust policy, and embedded all provided project evidence using portfolio documentation standard v1.0.1.

References

- Cloud Security Lab a Week (S.L.A.W.) - Create Your First AWS IAM Role.
- User-provided IAM role trusted entity and use-case screenshot.
- User-provided AmazonSSMMManagedInstanceCore permissions screenshot.
- User-provided role name, trust policy, and permissions summary screenshot.